Unit 2: OS Installation, Unix Shell, Commands & vi Editor

1. Basics of OS Installation & Prerequisites

Definition: Operating System (OS) installation is the process of setting up system software on a computer's hard drive so that it can manage hardware resources and provide an interface for user applications.

Before installing any operating system (like Windows or Linux), certain preparations are universally required to ensure a smooth setup without data loss.

Pre-Installation Requirements:

1. **ISO Image File:** An exact digital copy of the OS installation disc downloaded from the official website (e.g., Microsoft or Ubuntu).
2. **Bootable Media:** A USB flash drive (usually 8GB or more) formatted using tools like *Rufus* or *BalenaEtcher* to make it capable of booting the PC before the existing OS loads.
3. **BIOS/UEFI Configuration:** Accessing the motherboard's firmware (by pressing F2, F12, Del, or Esc during startup) to change the "Boot Order/Priority" so the computer reads the USB drive first instead of the Hard Drive.
4. **Backup Data:** Installing a new OS often involves formatting (erasing) the hard drive. All important data must be backed up externally.

👉 **Hinglish Explanation:** Computer mein naya OS (Windows ya Linux) dalne se pehle ek 'Bootable Pen Drive' banani padti hai jisme OS ka ISO file hota hai (Rufus software use karke). Phir computer restart karke BIOS settings mein jaate hain aur Boot menu mein pen drive ko first number par set karte hain taaki computer hard disk ki jagah pen drive se start ho. Apna data backup karna zaroori hai warna delete ho jayega.

2. MS Windows Installation (Step-by-Step)

Definition: Microsoft Windows is a widely used Graphical User Interface (GUI) based operating system. Installing it involves booting from media, partitioning the drive, and running the setup wizard.

Here is the standard procedure to install Windows 10/11 on a computer system:

Detailed Installation Steps:

1. **Boot from USB:** Insert the bootable Windows USB and start the PC. Press any key when prompted "Press any key to boot from USB...".
2. **Windows Setup Screen:** The installer loads. Select your *Language to install, Time and currency format, and Keyboard/input method*. Click **Next**, then click **Install Now**.
3. **Product Key & Edition:** Enter the 25-character product key if you have one, or click "I don't have a product key" to activate later. Select the OS edition (e.g., Windows 10 Pro) and accept the license terms.
4. **Installation Type:**
   - *Upgrade:* Keeps files, settings, and apps (used when updating an older Windows).
   - *Custom (Advanced):* Installs a fresh copy. Select this for a clean installation.
5. **Disk Partitioning:** The screen displays available hard drives. You can Delete old partitions, Format them, and click **New** to create a new partition. Windows uses the **NTFS** (New Technology File System) format. Select the primary partition and click Next.
6. **Copying & Installing:** The setup will now copy Windows files, expand them, install features, and install updates. The computer will restart automatically several times.
7. **OOBE (Out-Of-Box Experience):** After restarts, Cortana/Setup Wizard will ask for basic configurations: Connecting to Wi-Fi, setting up a Microsoft Account (or Offline account), creating a PIN/Password, and choosing privacy settings (Location, Diagnostics).
8. **Completion:** The system prepares the desktop and boots into the Windows Home Screen.

👉 **Hinglish Explanation:** Windows install karna bahut aasan hai. Bootable pen drive lagakar 'Custom Install' select karein. Iske baad hard disk ke purane partitions delete karke naya partition banayein (NTFS format me). 'Next' dabane par files copy hongi aur PC 2-3 baar restart hoga. Aakhir mein apna naam, password aur Wi-Fi connect karein, aur aapka naya Windows desktop samne aa jayega.

3. Linux OS Installation (e.g., Ubuntu)

Definition: Linux is an open-source, Unix-like operating system kernel. Distributions (Distros) like Ubuntu, CentOS, and Fedora package this kernel with GUI and software. Installation requires specific filesystem partitioning.

Installing Linux (using Ubuntu as the standard example) is slightly different from Windows, especially in how it handles file systems and partitions.

Detailed Installation Steps:

1. **Boot & Welcome Screen:** Boot from the Linux USB. The GRUB menu appears. Select "Install Ubuntu" (or "Try Ubuntu without installing" for a live environment).
2. **Keyboard & Network:** Select keyboard layout and connect to Wi-Fi (optional, but helps download updates during installation).
3. **Updates and Other Software:** Choose "Normal Installation" (includes web browser, utilities, office software, games) or "Minimal Installation" (only basics). Check the box "Install third-party software for graphics and Wi-Fi".
4. **Installation Type (Partitioning):** This is the most crucial step.
   - *Erase disk and install Ubuntu:* Wipes the entire hard drive automatically.
   - *Something else (Manual Partitioning):* Used for Dual-Booting with Windows. You must manually create at least two

partitions:

a. **Root Partition (/):** The main drive where the OS is installed. Format: **Ext4**.

b. **Swap Area:** Acts as virtual RAM. If physical RAM is full, the OS uses this space. Size is usually 1x to 2x of physical RAM.

5. **Location & User Info:** Select your timezone on the map. Enter your name, computer name, username, and a strong password. (This user gets 'sudo' privileges).
6. **File Copying & GRUB:** The installer copies files. At the end, it installs the **GRUB Bootloader**, which allows you to choose between Linux and Windows when the computer turns on.
7. **Restart:** Remove the installation media when prompted and hit Enter. The Linux desktop loads.

👉 **Hinglish Explanation:** Linux (Ubuntu) dalna Windows jaisa hi hai, par disk partition alag hota hai. Agar aapko manual partition ('Something else') karna hai, toh 2 cheezein zaroori hain: pehla 'Root (/)' partition jo Ext4 format me hota hai jaha OS rakha jayega. Dusra 'Swap area', jo tab kaam aata hai jab aapki RAM full ho jati hai (ye virtual RAM ka kaam karta hai). Install hone ke baad GRUB bootloader dalta hai jisse PC on hone par OS select karne ka option aata hai.

4. Unix Shell and Its Architecture

Definition: A Unix Shell is a command-line interpreter (CLI) that acts as the primary interface between the user and the Unix/Linux operating system. It takes commands typed by the user, translates them, and passes them to the Kernel for execution.

Unix Operating System Architecture:

The Unix OS is structured in layers, often visualized as concentric circles:

- **Hardware:** The physical components (CPU, RAM, Disks).
- **Kernel:** The core of the OS. It interacts directly with the hardware, managing memory, processes, and file systems.
- **Shell:** The outer layer wrapping the Kernel. It protects the user from complex Kernel language. It reads human-readable commands, parses them, and tells the Kernel what to do.
- **Applications/Utilities:** Programs running on the OS (Browsers, Editors, Compilers).

```
+----------------------------------------+
|         Applications / Users           |
|   +--------------------------------+   |
|   |           SHELL                |   |
|   |   +------------------------+   |   |
|   |   |        KERNEL          |   |   |
|   |   |   +----------------+   |   |   |
|   |   |   |   HARDWARE     |   |   |   |
|   |   |   +----------------+   |   |   |
|   |   +------------------------+   |   |
|   +--------------------------------+   |
+----------------------------------------+
```

Types of Unix Shells:

Over the years, various shells have been developed with different features and scripting syntaxes:

1. **Bourne Shell (sh):** The original Unix shell. Prompt symbol is `$`. It lacks modern features like command history but is highly standardized.
2. **Bash (Bourne Again Shell):** The default shell for most modern Linux distributions (Ubuntu, Kali). It is an upgraded version of `sh` with command history, auto-completion, and better scripting.
3. **C Shell (csh):** Its syntax resembles the C programming language. Prompt symbol is `%`.
4. **Korn Shell (ksh):** Combines the best features of C Shell and Bourne Shell. Highly efficient for scripting.

> 👉 **Hinglish Explanation:** Computer ka hardware direct hamari bhasha nahi samajhta. **Kernel** OS ka boss hai jo hardware chalata hai. Par hum Kernel se direct baat nahi kar sakte. Isliye hum **Shell** (jaise Bash ya Terminal) ka use karte hain. Hum Shell me command type karte hain, Shell usko translate karke Kernel ko deta hai, aur Kernel hardware se kaam karwa ke output wapas Shell ke zariye hume dikhata hai.

5. Basic Unix / Linux Commands

Definition: Unix commands are text-based instructions typed into the shell to perform specific tasks such as file manipulation, directory navigation, system monitoring, and permission changes.

Below are the most fundamental and frequently used commands that every IT student must know. Unix is **case-sensitive**, meaning `Ls` and `ls` are treated differently.

Command	Description	Example / Syntax
pwd	Print Working Directory. Shows the exact absolute path of the current folder you are in.	**pwd** (Output: /home/user/documents)
ls	List directory contents. Shows files and folders in the current location.	**ls** (basic list) **ls -l** (detailed long list)
cd	Change Directory. Used to navigate between folders.	**cd /etc** (go to /etc) **cd ..** (go back one folder)
mkdir	Make Directory. Creates a new empty folder.	**mkdir myfolder**
rmdir	Remove Directory. Deletes an *empty* folder.	**rmdir myfolder**
touch	Creates a new empty file, or updates the timestamp of an existing file.	**touch file1.txt**
cp	Copy files or directories from source to destination.	**cp file1.txt /home/backup/**
mv	Move (cut/paste) or Rename a file or directory.	**mv oldname.txt newname.txt** (renames)
rm	Remove (delete) a file permanently. No recycle bin!	**rm file1.txt** **rm -rf folder/** (force delete folder)
cat	Concatenate. Used to display the entire content of a file on the screen.	**cat notes.txt**

chmod	Change Mode. Changes read, write, and execute permissions of a file.	**chmod 777 file.sh** (gives all permissions)
grep	Global Regular Expression Print. Searches for a specific word/pattern inside a file.	**grep "error" log.txt**

👉 **Hinglish Explanation:** Bina mouse ke computer chalane ke liye commands yaad honi chahiye. Exam mein kisi bhi command ka naam dekar puchenge, toh aapko uska kaam aur ek example likhna hoga. Jaise `mkdir` (Make Directory) folder banata hai, `cd` (Change Directory) folder badalta hai, aur `rm` file delete karta hai. Linux me sab commands chhote aksharon (lowercase) me hoti hain.

6. The 'vi' Editor (Visual Editor)

Definition: 'vi' (Visual Editor) is a powerful, screen-oriented text editor created for the Unix operating system. It comes pre-installed on almost every Linux/Unix system and operates entirely via the keyboard without a mouse.

Unlike modern graphical editors (like Notepad or MS Word), opening the vi editor drops you into an interface where every key press is a command until you explicitly tell it you want to type text. It operates on the concept of **Modes**.

Three Modes of vi Editor:

1. **Command Mode (Default Mode):** When you open a file using `vi filename`, it opens in this mode. Here, keys act as commands (to copy, paste, delete, navigate) rather than typing letters. You cannot type words in this mode.
2. **Insert Mode:** This is the mode where you can actually type and edit text. You enter this mode by pressing specific keys (like `i` or `a`) from the command mode. To exit Insert mode and return to Command mode, you must press the **ESC** key.
3. **EX Mode (Last Line Mode):** Used for saving the file, exiting the editor, or searching. Accessed by pressing the colon `:` key while in Command mode. The cursor drops to the bottom left of the screen.

```
MODE SWITCHING DIAGRAM:

                    [Press 'i', 'a', 'o']
                 -------------------------->
[COMMAND MODE]                                [INSERT MODE]
 (Default)       <--------------------------   (Typing Text)
                     [Press 'ESC' key]
      | ^
      | |
[:/ ?]| | [Press 'Enter']
      v |
 [EX / LAST LINE MODE]
 (Saving, Quitting)
```

Important vi Commands:

To Enter Insert Mode:

- **i** : Insert text *before* the current cursor position.
- **a** : Append text *after* the current cursor position.
- **o** : Open a new line *below* the current line and enter insert mode.

In Command Mode (Navigation & Editing):

- **x** : Delete the single character under the cursor.
- **dd** : Delete (cut) the entire current line.
- **yy** : Yank (copy) the entire current line.
- **p** : Paste the copied or deleted line below the cursor.
- **u** : Undo the last change.

In EX Mode (Saving & Exiting):

- **:w** : Write (Save) the file but don't exit.
- **:q** : Quit the editor (fails if there are unsaved changes).
- **:q!** : Force quit WITHOUT saving changes.
- **:wq** or **:x** : Save changes and Quit safely.

👉 **Hinglish Explanation:** 'vi' editor Linux ka Notepad hai, bas problem ye hai ki isme mouse nahi chalta aur khulte hi aap type nahi kar sakte. Khulte hi ye **Command Mode** me hota hai. Likhna shuru karne ke liye pehle **'i'** (insert) dabana padta hai. Likhne ke baad agar save karna hai, toh pehle **ESC** button dabao (wapas command mode me aane ke liye), phir colon **:wq** likh kar Enter maro (w matlab write/save, q matlab quit). Ek line delete karne ke liye 'dd' dabate hain.

🔁 Quick Revision Notes (Exam Saver)

- **OS Installation Basics:** Needs ISO image, Bootable USB (Rufus), BIOS boot priority set to USB.
- **Linux Partitions:** Crucial partitions during manual install are Root (/) as Ext4 and Swap Area (Virtual RAM).
- **Unix Architecture:** Hardware → Kernel (core manager) → Shell (translator/interface) → Application.
- **Types of Shells:** Bourne (sh), Bash (default in Ubuntu), C-shell (csh), Korn (ksh).
- **Key Commands:** `pwd` (current path), `ls` (list), `mkdir` (make folder), `cp` (copy), `mv` (move/rename), `rm` (delete).
- **vi Editor Modes:** Command mode (default, for copy/delete), Insert mode (press 'i' to type text), Ex mode (press ':' to save/quit).
- **vi Shortcuts:** `:wq` (save & quit), `dd` (delete line), `yy` (copy line), `ESC` (exit insert mode).

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